

DOCUMENT CONTROL BLOCK

Title: Assembly Line Sensor Integration & Digital Twin Sync

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MEETING NOTES FROM 10/10/2025

Before we get into the SOP, a quick reminder that the breakroom fridge on floor 3 is being cleaned out this Friday. Anything not labeled will be thrown away. Also, parking spaces 44-50 are reserved for the executive team next week during the audit.

PROCESS OVERVIEW

This document outlines how we connect physical assembly line hardware to our virtual models. It is critical that this is followed strictly to ensure data integrity.

STEP-BY-STEP PROCEDURE

When a Hardware Engineer physically installs a new Industrial IoT sensor on the factory floor, they must immediately write down the MAC address on a physical clipboard. At the end of their shift, the Hardware Engineer manually types these MAC addresses into the legacy ERP system to register the device.

Once registered, the sensor begins transmitting data. However, this raw data is often noisy. Therefore, the Data Analytics team is required to intercept this feed. A Data Analyst will run a custom Python script to scrub and validate the sensor output for a 24-hour observation period.

If the Python script flags the data as clean, the Data Analyst logs into the admin dashboard and manually approves the feed. Only after this human approval does the telemetry route to the main Java Full Stack portal, which stores the records permanently in our PostgreSQL database.

Finally, we need to keep our simulation updated. Every Friday at 5:00 PM, the System Admin exports a CSV batch file from the PostgreSQL database. The System Admin then manually imports this CSV batch file into the Unity Game Engine to update the factory's digital twin for the weekly Monday review meeting.

NOTE: Do not use the new Autodesk platform for any of this yet, the licenses are still pending approval from finance.